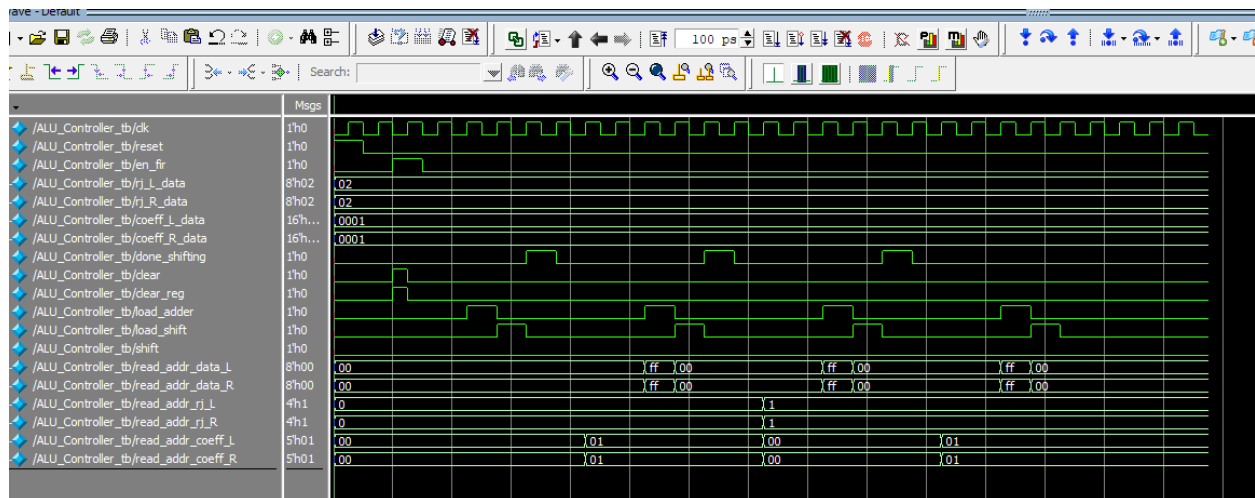


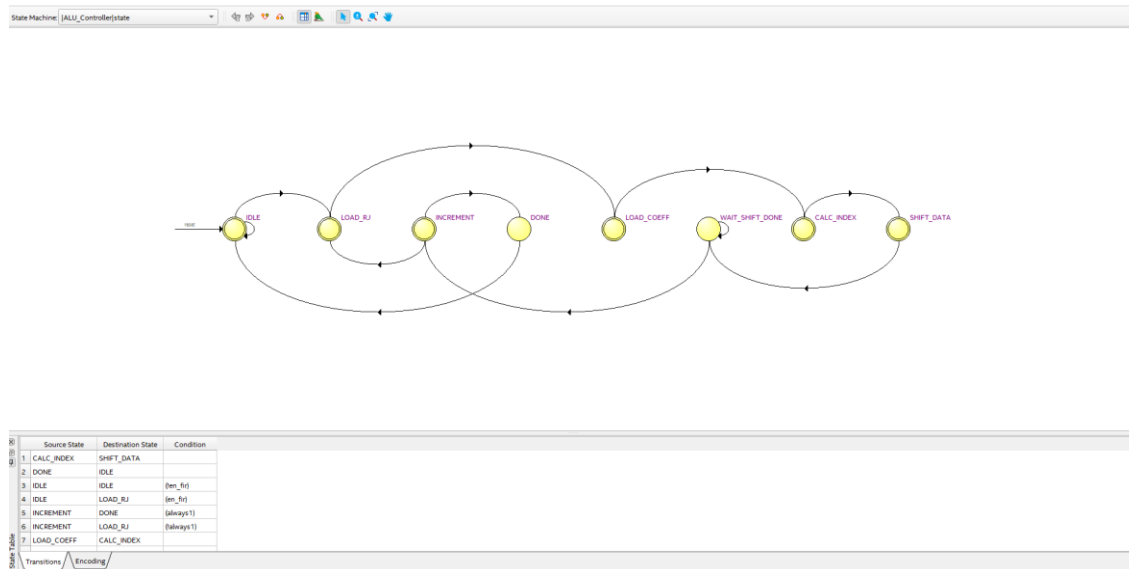
ALU_Controller Block

This controls the interaction between the ALU and other Blocks inside the MSDAP. It also controls the local behavior of the ALU block meaning how the internal blocks inside the ALU interact with each other in terms of shifting and computation. The main challenge in making this block is figuring out when to bring in datain sample or which address of the data_memory to use. An index was created to compute the address of the data_memory to be used at each stage of the convolution. This block is the most challenging part of the ALU block.

WaveView after the Simulation



State Diagram of the ALU Finite State Machine



ALU Block diagram: the yellow is the FSM block of the ALU controller

